

$$Z_1 = 6000$$

$$P_1 = 300$$

$$P_2 = 40000$$

$$\omega_0 = 1$$

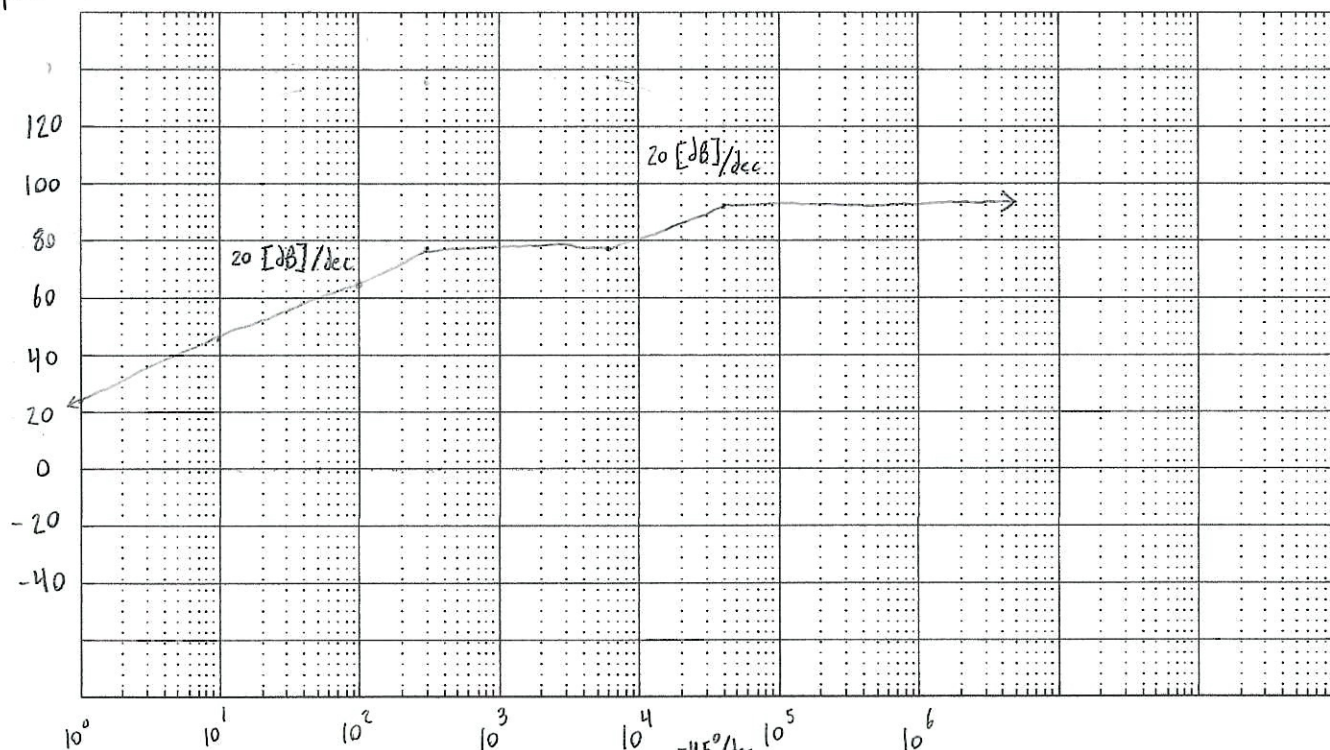
$$H(j1) = \frac{-j20}{1} \Rightarrow$$

$$20 \log(H(j1)) = 26 \text{ [dB]}$$

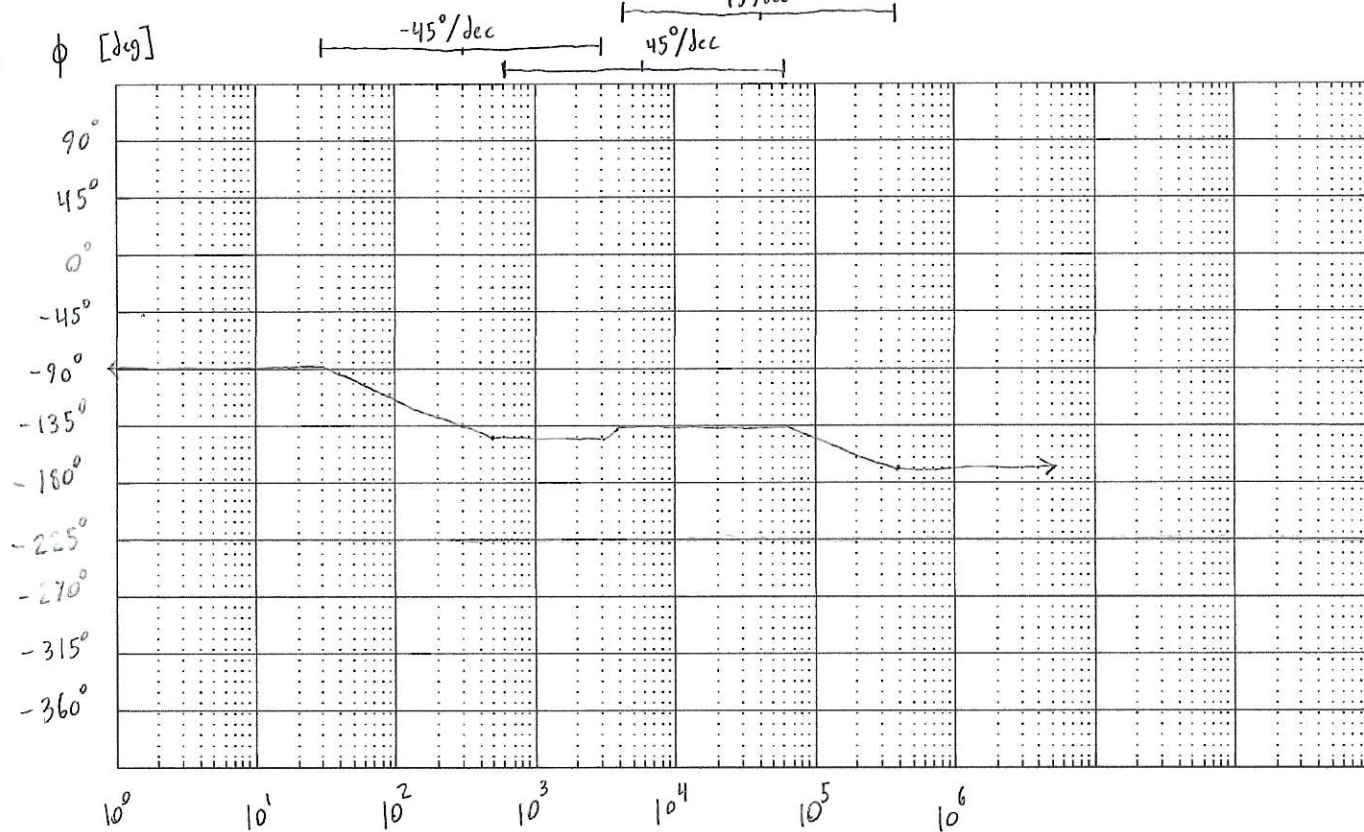
$$\phi = -90^\circ$$

$$1.)$$

$$|H(j\omega)| \text{ [dB]}$$



$$\phi \text{ [deg]}$$



2.)

$$\omega_0 = 30$$

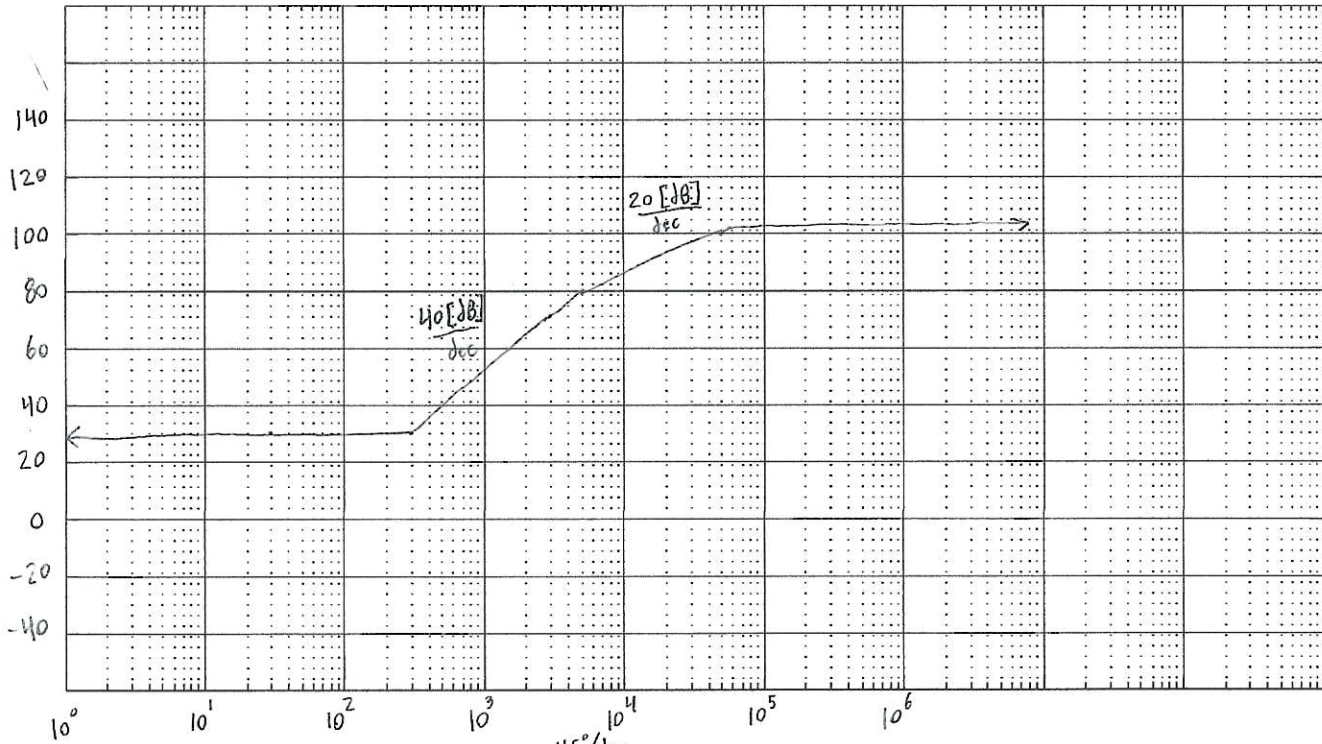
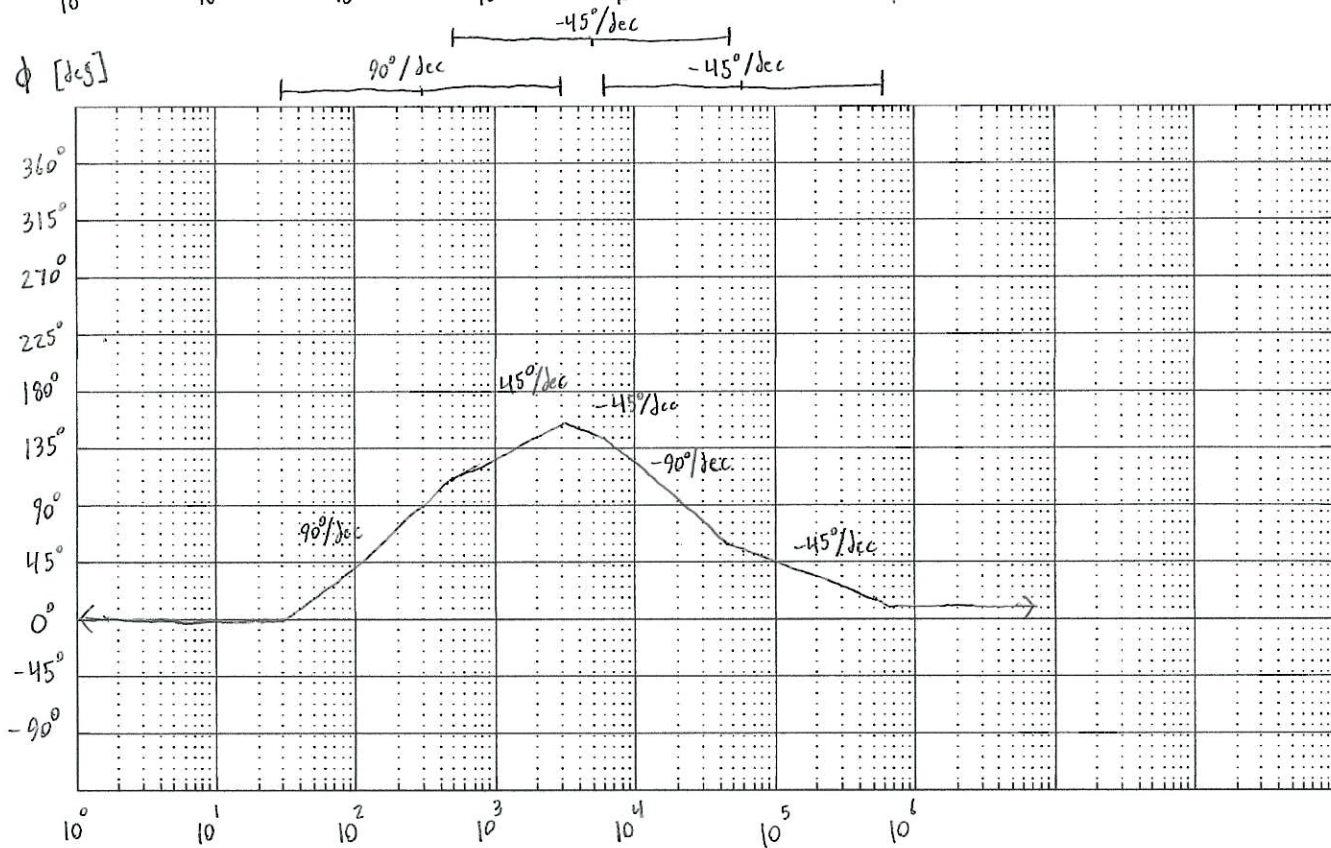
$$H(j30) = -30 \Rightarrow 29.5 \text{ [dB]}$$

$$\phi = 0^\circ$$

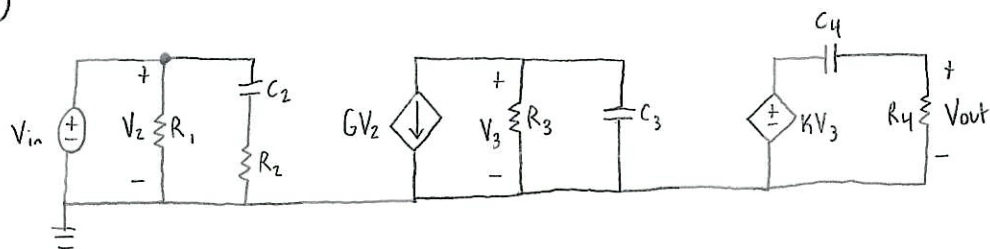
$$Z_1 = Z_2 = 300$$

$$p_1 = 5000$$

$$p_2 = 60000$$

 $|H(j\omega)| \text{ [dB]}$

 $\phi \text{ [deg]}$


3.) a.)



$$V_2 = V_{in}$$

$$\text{KCL: } GV_2 + \frac{V_3}{R_3} + \frac{V_3}{(1/j\omega C_3)} = 0$$

$$(V_3) \quad GV_2 + V_3 \left(\frac{1}{R_3} + \frac{j\omega C_3}{1} \right) = 0$$

$$GV_2 + V_3 \left[\frac{1 + j\omega C_3 R_3}{R_3} \right] = 0$$

$$GV_2 + V_3 \left[\frac{1 + j\omega C_3 R_3}{R_3} \right] = 0$$

$$V_3 = -GV_2 \cdot \frac{R_3}{1 + j\omega C_3 R_3}$$

$$V_{out} = \frac{R_4}{1/j\omega C_4 + R_4} KV_3$$

$$= \frac{j\omega C_4 R_4}{1 + j\omega C_4 R_4} KV_3$$

$$= -GV_2 K R_3 \frac{j\omega C_4 R_4}{(1 + j\omega C_3 R_3)(1 + j\omega C_4 R_4)}$$

$$\frac{V_{out}}{V_{in}} = -GR_3K \cdot \frac{j\omega C_4 R_4}{(1 + j\omega C_3 R_3)(1 + j\omega C_4 R_4)}$$

b.)

$$H(j\omega) = -10 \cdot 10 \cdot \frac{j\omega 0.01}{(1 + j\omega 0.0001)(1 + j\omega 0.01)}$$

$$H(j\omega) = -100 \cdot \frac{j \frac{\omega}{100}}{(1 + j \frac{\omega}{10000})(1 + j \frac{\omega}{100})}$$

See next page for Bode plots

3.) b.)

$$\omega_0 = 10$$

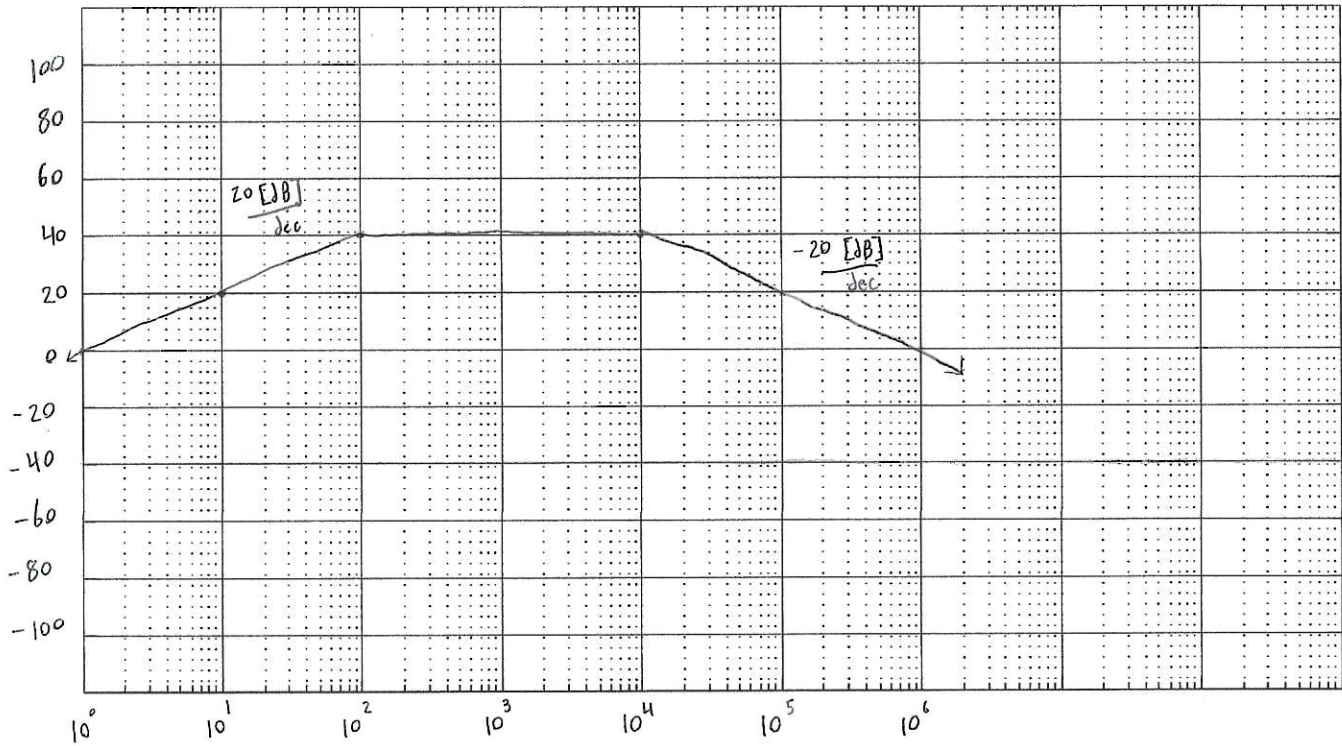
$$H(j\omega) = -j10 \Rightarrow 20 \text{ [dB]}$$

$$\phi = -90^\circ$$

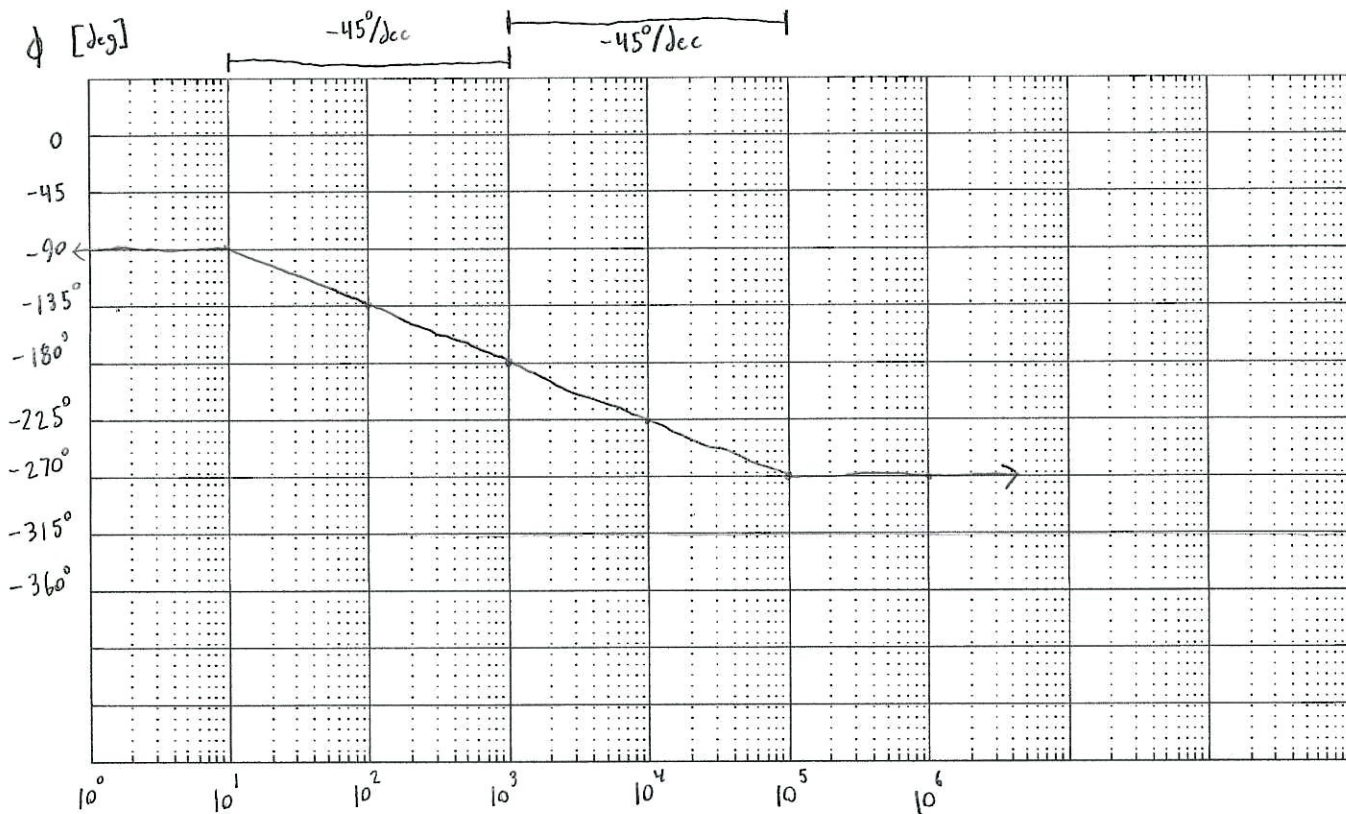
$$p_1 = 100$$

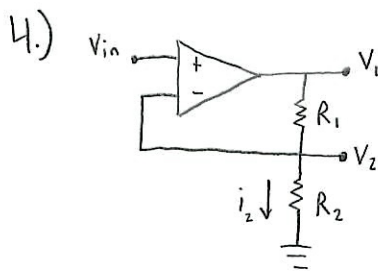
$$p_2 = 10000$$

$|H(j\omega)| \text{ [dB]}$



$\phi \text{ [deg]}$



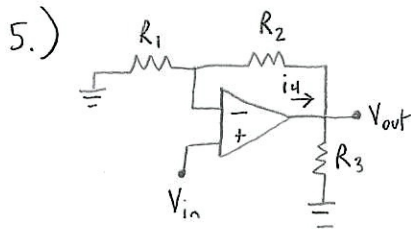


$$V_2 = V_{in}$$

$$i_2 = \frac{V_{in}}{R_2}$$

$$V_1 = V_{in} + i_2 R_1 = V_{in} + V_{in} \frac{R_1}{R_2}$$

$$V_1 = V_{in} \left(1 + \frac{R_1}{R_2} \right)$$



KCL: $\frac{-V_{in}}{R_1} = \frac{V_{in} - V_{out}}{R_2}$

$$\frac{V_{out}}{R_2} = \frac{V_{in}}{R_1} + \frac{V_{in}}{R_2}$$

$$V_{out} = \left(1 + \frac{R_2}{R_1} \right) V_{in}$$

$$i_4 = \frac{V_{out}}{R_3} + \frac{V_{out} - V_{in}}{R_2}$$

$$= V_{out} \left(\frac{1}{R_3} + \frac{1}{R_2} \right) - \frac{V_{in}}{R_2}$$

$$= \left(1 + \frac{R_2}{R_1} \right) \left(\frac{1}{R_3} + \frac{1}{R_2} \right) V_{in} - \frac{V_{in}}{R_2}$$

$$= \left(\frac{1}{R_3} + \frac{1}{R_2} + \frac{R_2}{R_1 R_3} + \frac{1}{R_1} \right) V_{in} - \frac{V_{in}}{R_2}$$

$$i_4 = \left(\frac{1}{R_1} + \frac{1}{R_3} + \frac{R_2}{R_1 R_3} \right) V_{in}$$